

SciDaVinci demo background: cytokine box plot

Experiment overview:

This synthetic demo describes an in vitro macrophage inflammation assay.

Cells were stimulated with LPS for 24 hours, then treated with candidate drugs.

The table summarizes IL-6 secretion in ng/mL for each condition.

Groups:

Control: LPS stimulation without drug treatment.

Low-dose Drug A: LPS plus a low dose of Drug A.

High-dose Drug A: LPS plus a high dose of Drug A.

Drug B: LPS plus a standard dose of a positive-control drug.

Biological context:

IL-6 is a pro-inflammatory cytokine. Higher values usually indicate a stronger inflammatory response.

Drug A is a candidate immunomodulator. Early pilot observations suggest high-dose Drug A may trigger cellular stress.

Drug B is expected to suppress or maintain low IL-6 levels relative to strong inflammatory activation.

How to interpret clicked chart elements:

A clicked High-dose Drug A box with a higher median and wider IQR suggests stronger and more variable IL-6 induction.

Clicked High-dose Drug A outliers may represent samples with unusually strong stress or inflammatory activation.

A clicked Drug B box near or below Control supports the expected positive-control behavior.

A clicked Control box provides the LPS-only baseline for comparing all treated groups.

Suggested user prompt:

Use box_cytokine_summary.csv and this PDF background. Create exactly one interactive box plot.

Then, when I click a chart element, explain the selected element using the biological background.